Ex. No: 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	08.09.2026
Git-hub Host Link	https://fr8st7.github.io/flutter_exp_5/

Aim:

To implement navigation between multiple Flutter screens using Navigator and named routes.

Description:

This experiment demonstrates how to navigate between multiple screens in a Flutter application. The application contains **Home, Profile, and Details** screens. Named routes are used for navigation, and Navigator.pop() is used to return to the previous screen. A colorful **purple, pink, and sky-blue** theme is used to improve the user interface.

Procedure:

Step 1: Create a new Flutter project using VS Code or the terminal.

Step 2: Create separate Dart files for the Home, Profile, and Details screens inside the lib folder.

Step 3: Design the Home screen with a button to navigate to the Profile screen.

Step 4: Design the Profile screen with a button to navigate to the Details screen.

Step 5: Design the Details screen with a Back button to return to the previous screen.

Step 6: Define named routes for Home, Profile, and Details inside MaterialApp.

Step 7: Use Navigator.pushNamed() to navigate between the screens.

Step 8: Use Navigator.pop() to return to the previous screen.

Step 9: Apply a colorful user interface using purple, pink, and sky-blue colors, gradients, cards, icons, and buttons.

Step 10: Run the application using flutter run and check the navigation between all screens.

Step 11: Verify the navigation flow: **Home → Profile → Details** and back to the previous screens.

Step 12: Confirm that navigation and routing work correctly in the Flutter application.

Program:

main.dart -

```
import 'package:flutter/material.dart';
import 'home.dart';
import 'profile.dart';
import 'details.dart';

void main() {
  runApp(const NavigationApp());
}

class NavigationApp extends StatelessWidget {
  const NavigationApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Navigation App',

      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: Colors.purple,
        ),
        useMaterial3: true,
      ),

      initialRoute: '/',

      routes: {
        '/': (context) => const HomePage(),
        '/profile': (context) => const ProfilePage(),
        '/details': (context) => const DetailsPage(),
      },
    );
  }
}
```

home.dart -

```
import 'package:flutter/material.dart';
class HomePage extends StatelessWidget {
const HomePage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar:  AppBar(
        title: const Text(
        'Home',
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
        centerTitle: true,
        backgroundColor: Colors.purple,
      ),
      body: Container(
        decoration: const BoxDecoration(
          gradient: LinearGradient(
          colors: [
              Colors.purple,
              Colors.pink,
              Colors.lightBlue,
            ],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ),
        ),
        child: Center(
          child: Card(elevation: 10,
            margin: const EdgeInsets.all(30),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(25),
            ),
            child: Padding(
              padding: const EdgeInsets.all(35),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.home_rounded,
                    size: 80,
```

```
  color: Colors.purple,
),
const SizedBox(height: 20),
const Text(
  'Welcome Home!',
  style: TextStyle(
  fontSize: 28,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 10),
const Text(
  'Explore the different screens',
  style: TextStyle(
    fontSize: 16,
    color: Colors.grey,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon(
onPressed: () {
    Navigator.pushNamed(context, '/profile');
  },
  icon: const Icon(Icons.person),
  label: const Text('Go to Profile'),
  style: ElevatedButton.styleFrom(
  backgroundColor: Colors.pink,
  foregroundColor: Colors.white,
    padding: const EdgeInsets.symmetric(
      horizontal: 25,
      vertical: 15,
    ),
    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(30),
       ),),),],),),),),),);}}
```

profile.dart -

```
import 'package:flutter/material.dart';
class ProfilePage extends StatelessWidget {
  const ProfilePage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar:  AppBar(
        title: const Text(
        'Profile',
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
        centerTitle: true,
        backgroundColor: Colors.pink,
      ),
      body: Container(
        decoration: const BoxDecoration(
          gradient: LinearGradient(
          colors: [
              Colors.pink,
              Colors.purple,
              Colors.lightBlue,
            ],
            begin: Alignment.topRight,
            end: Alignment.bottomLeft,
          ),
        ),
        child: Center(
          child: Card(
          elevation: 10,
            margin: const EdgeInsets.all(30),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(25),
            ),
            child: Padding(
              padding: const EdgeInsets.all(35),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const
```

```
CircleAvatar(
  radius: 50,
  backgroundColor: Colors.pink,
  child: Icon(Icons.person,

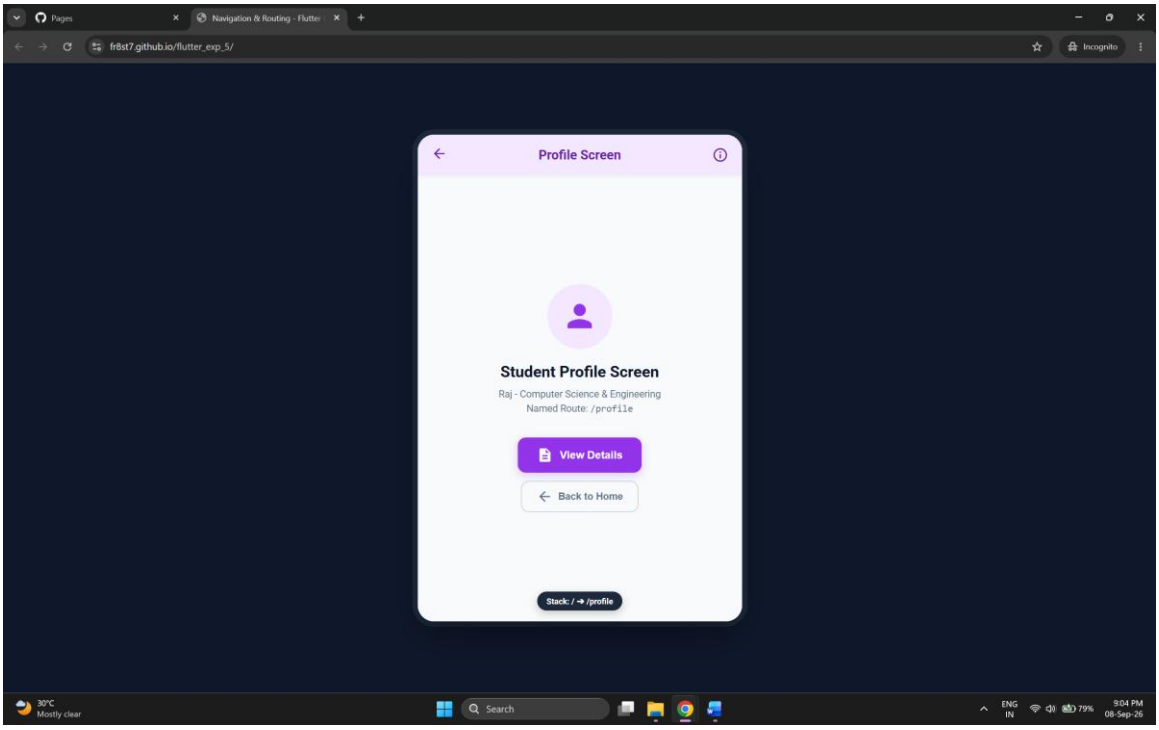
    size: 60,
    color: Colors.white,
  ),
),
const SizedBox(height: 20),
const Text('Student Profile',
style: TextStyle(
    fontSize: 27,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 10),
const Text('Welcome to your profile page!',
  style: TextStyle(
    fontSize: 16,
    color: Colors.grey,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon(
onPressed: () {
    Navigator.pushNamed(context, '/details');
  },
  icon: const Icon(Icons.info_outline),
  label: const Text('View Details'),
  style: ElevatedButton.styleFrom(
  backgroundColor: Colors.lightBlue,
  foregroundColor: Colors.white,
    padding: const EdgeInsets.symmetric(
      horizontal: 25,
      vertical: 15,
    ),
    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(30),
       ),),),],),),),),),);}
```

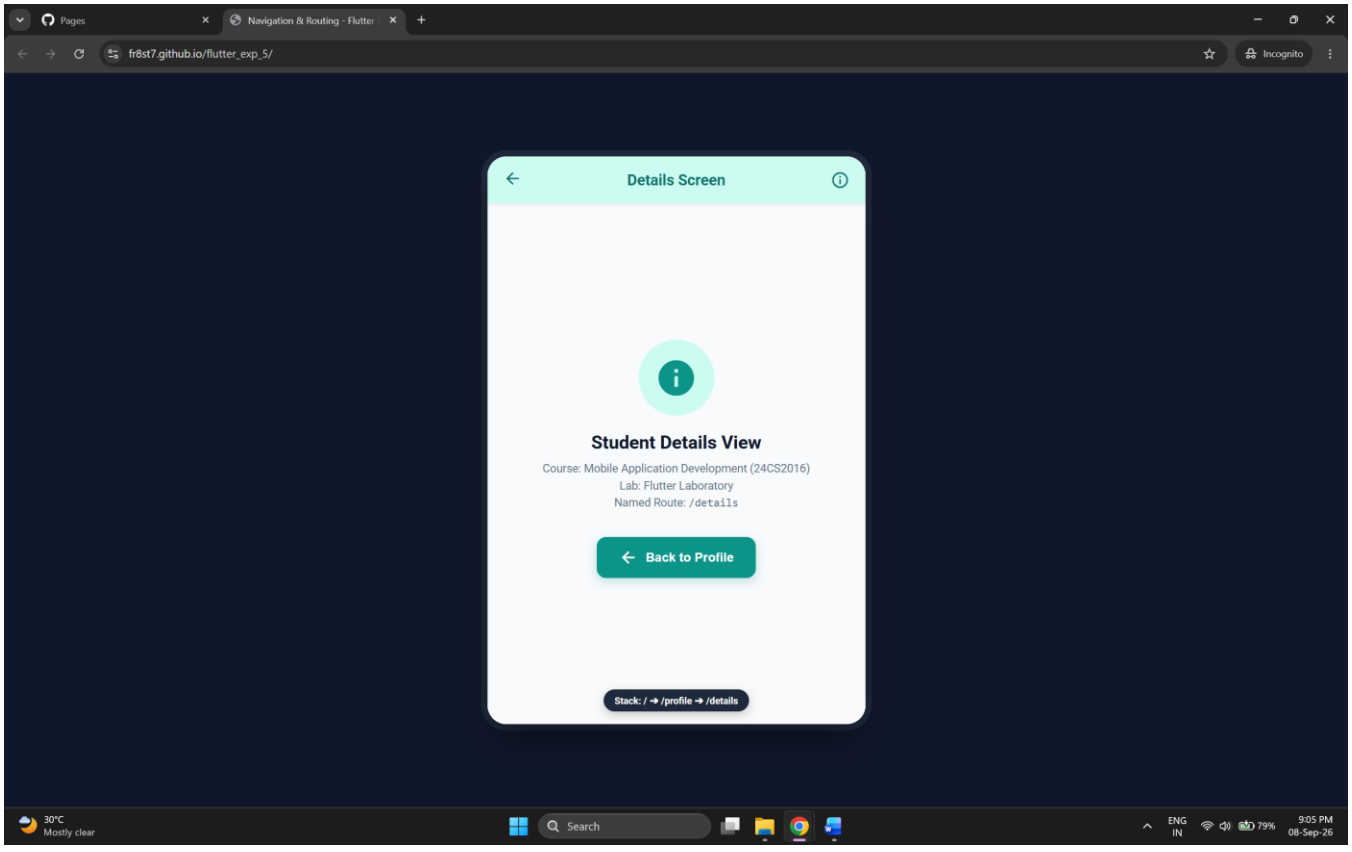
details.dart -

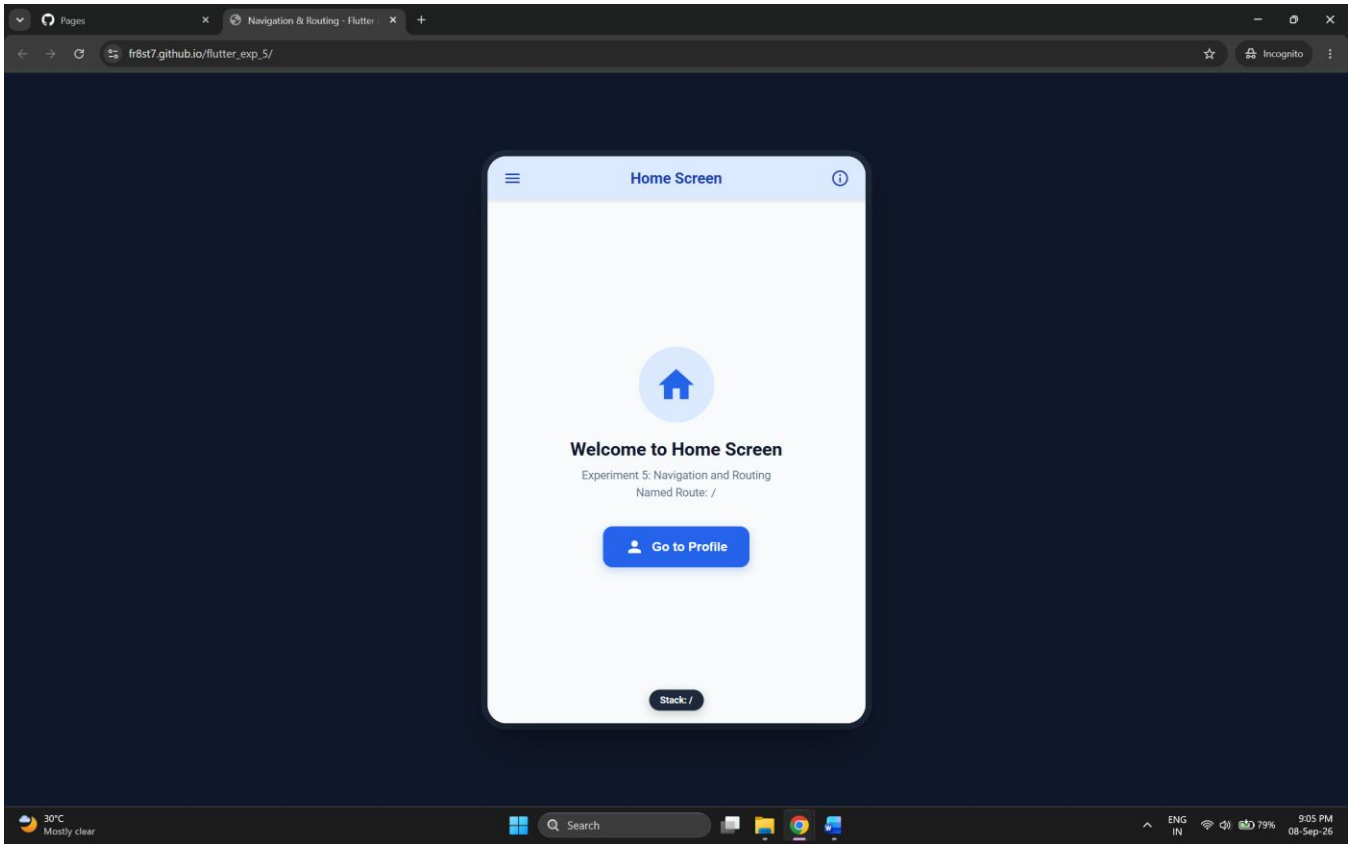
```
import 'package:flutter/material.dart';
class DetailsPage extends StatelessWidget {
  const DetailsPage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar:  AppBar(
        title: const Text(
        'Details',
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
        centerTitle: true,
        backgroundColor: Colors.lightBlue,
      ),
      body: Container(
        decoration: const BoxDecoration(
          gradient: LinearGradient(
          colors: [
              Colors.lightBlue,
              Colors.purple,
              Colors.pink,
            ],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ),
        ),
        child: Center(
          child: Card(
          elevation: 10,
            margin: const EdgeInsets.all(30),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(25),
            ),
            child: Padding(
              padding: const EdgeInsets.all(35),
              child: Column(mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.school_roun
```

```
  color: Colors.lightBlue,
),
const SizedBox(height: 20),
const Text('Student Details',
style: TextStyle(
    fontSize: 27,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 20),
const Text(
  'Name: Arsath Ahamed URK23CS1203\n'
  'Course: Computer Science\n'
  'Department: CSE',
  textAlign: TextAlign.center,
  style: TextStyle(
    fontSize: 17,
    height: 1.7,
    color: Colors.black87,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon(onPressed: () {
    Navigator.pop(context);
  },
  icon: const Icon(Icons.arrow_back),
  label: const Text('Back to Profile'),
  style: ElevatedButton.styleFrom(
  backgroundColor: Colors.purple,
  foregroundColor: Colors.white,
  padding: const EdgeInsets.symmetric(
      horizontal: 25,
      vertical: 15,
    ),
    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(30),
    ),),),],),),),),),);}}
```

Output:


Profile Screen
Student Profile Screen
Raj - Computer Science & Engineering
Named Route: /profile
View Details
Back to Home
Stack: / → /profile


Details Screen
Student Details View
Course: Mobile Application Development (24CS2016)
Lab: Flutter Laboratory
Named Route: /details
Back to Profile
Stack: / → /profile → /details



Result:

The Flutter application was successfully implemented with navigation between Home, Profile, and Details screens using named routes and back navigation.